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The signaling effect of tone: The influence of key audit matters' tone on bank lending decisions

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ABSTRACT

I apply the BERT model to assess the tone of key audit matters (KAMs) disclosed by auditors of Chinese A-share companies from 2016 to 2022, and investigate the impact of tone on bank lending decisions. The findings indicate that a negative tone serves as a risk signal by helping reveal information risk and assess credit risk. Positive and negative tones exhibit asymmetric signaling effects, with expected tones exerting a stronger influence than unexpected tones. These results highlight that the audit report's tone is a more credible risk indicator than management's tone, thereby providing strong support for banks' credit decision-making process.

1. Introduction

Global regulators have recently promoted audit report reforms to enhance the informational value and transparency of audit reporting (Lennox et al., 2023). In 2016, China introduced the "Chinese Auditing Standards No. 1504—Communicating Key Audit Matters in the Audit Report" (hereinafter referred to as "Auditing Standard No. 1504"), mandating the disclosure of key audit matters (KAMs). KAMs refer to substantial risks of material misstatements and reflect auditors' most complex, subjective, and challenging judgments (Reid et al., 2019; Lennox et al., 2023). KAMs' tone serves as a critical channel for conveying risk information and supplementing audit opinions, thereby helping stakeholders better understand the risks embedded in financial statements (Seebeck and Kaya, 2023; Smith, 2023; Yao et al., 2023). While previous studies mainly focus on KAMs' tone in equity markets (Seebeck and Kaya, 2023), its role in debt markets, such as bank lending, remains underexplored. Banks play critical roles in corporate financing worldwide (Graham et al., 2008; Tang et al., 2024). Therefore, understanding KAMs' tone within the context of debt markets is essential. In emerging markets, where information asymmetry tends to be greater, banks rely more heavily on audit reports than equity investors, particularly with respect to risk disclosure (Ball et al., 2008; Hasan et al., 2014; Porumb et al., 2021). Drawing on signaling theory, KAMs' tone can be viewed as an important signal conveyed by auditors to banks regarding a company's risk in environments characterized by information asymmetry. The central question is whether KAMs' tone effectively signals risk to banks and influences their lending decisions. Addressing this question will bridge the gap in understanding the role of KAMs' tone in debt markets while offering practical insights for auditors, banks, and regulators seeking to utilize KAMs' tone for effective risk assessment.

I select China as the research context for two reasons. First, compared with developed markets such as the United Kingdom, France, and the United States, emerging markets such as China have distinct institutional environments. In developed markets with strict regulations and comprehensive disclosures, the incremental information provided by KAMs is relatively limited (Gutierrez et al., 2018; Bédard et al., 2019; Burke et al., 2023). In contrast, China's regulatory framework continues to evolve, and its voluntary disclosures have considerable room for improvement (Li and Zheng, 2024). KAMs play a critical role in bridging information gaps and enhancing

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communication regarding key risk factors. Second, China's financial system is predominantly bank-based, with a high proportion of small- and medium-sized enterprises (SMEs) (Tang et al., 2024). Although these companies rely heavily on bank loans, they face particularly severe information asymmetry. In this context, banks rely on audit information, such as KAMs, to assess risks (Porumb et al., 2021). My study provides valuable insights for emerging markets such as China and offers practical implications for developed markets as KAMs are implemented globally and bank loans remain a critical component of corporate financing.

Using a sample of A-share listed companies on the Shanghai and Shenzhen stock exchanges from 2016 to 2022, I employ the BERT language model to assess KAMs' tone and examine its impact on bank lending terms. I focus on how a negative tone serves as a risk signal by helping reveal information risk and assess credit risk. Additionally, I explore how different types of tone distinctly influence bank lending terms, emphasizing the nuances of tone in communicating information risk. My findings reveal that a 1-standard deviation increase in KAMs' negative tone is associated with a 3.002 % increase in loan interest rates, a 7.225 % decrease in loan amounts, and an 8.485 % reduction in loan terms. The probability of requiring collateral increases by 71.600 % under a strongly negative tone compared with a weakly negative tone. These results extend the work of Ertugrul et al. (2017) and Chen et al. (2016), who report that the tone of annual reports and modified audit opinions affect lending terms. Furthermore, lenders prioritize KAMs' tone over their quantity or nature when assessing corporate financial risk, complementing Porumb et al.'s (2021) findings. I demonstrate that KAMs' tone has more substantial explanatory power in emerging markets such as China, where information asymmetry is more pronounced.

My study's contributions are threefold. First, my study broadens the scope and methodology of the research on KAMs' tone. Previous studies typically rely on the Loughran and McDonald (2011) dictionary to assess KAMs' tone in equity markets (Seebeck and Kaya, 2023). I employ the BERT model, which accounts for word order and context (Huang et al., 2023), enabling more accurate tone detection and signaling in debt markets. In doing so, I extend the research on KAMs' tone and improve identification methods. Second, I expand the boundaries of financial text-tone research. Most studies focus on the tone of annual reports (Dang et al., 2022; Ertugrul et al., 2017; Guo and Huang, 2024; Wei et al., 2024; Zhou et al., 2024), earnings calls (Dang et al., 2024), and analyst reports (Liang et al., 2022). Ertugrul et al. (2017) find that the annual report's tone affects loan terms. I demonstrate that KAMs' tone is a stronger risk signal than that of management reports, extending Ertugrul et al. (2017) and confirming prior experimental findings (Christensen et al., 2014; Sirois et al., 2018). Finally, I enhance the research on the role of audit reports in lending decisions. While prior studies focus on audit opinions and the number of KAMs in debt markets (Chen et al., 2016; Porumb et al., 2021), few analyze the language of audit texts (Bédard et al., 2016). By controlling for these factors, my study reveals that KAMs' tone significantly influences lending decisions, advancing research on audit report elements in debt markets.

2. Theoretical analysis and hypothesis development

The signaling theory suggests that credit market risks arise from information asymmetry between banks and firms (Stiglitz and Weiss, 1981). To make credit decisions, banks evaluate a firm's information and credit risk using operational data. However, when firms have an informational advantage, they may manipulate information to influence decisions, leading to adverse selection and moral hazard. As information intermediaries, auditors reduce credit risk by improving the quality of information disclosure (DeFond and Zhang, 2014). Audit reports are essential tools for communicating information risk, with audit opinions and KAMs serving as key references for banks in identifying information risk in publicly traded firms (Chen et al., 2016; Porumb et al., 2021). However, the uniformity of audit opinions and KAMs limits their ability to reflect differences in auditor behavior, restricting their information-transmission function (Lennox et al., 2023; Seebeck, 2024). Conversely, auditors have discretion over language use and can convey variations in risk perception through the tone of KAMs, helping banks in credit decision-making (Seebeck and Kaya, 2023; Smith, 2023).

KAMs' negative tone plays a crucial role in signal transmission by helping reveal information risk. KAMs often highlight areas with a high risk of material misstatements, such as revenue recognition (Reid et al., 2019). Specifically, a negative tone indicates that the auditor holds negative expectations or perceives greater uncertainty regarding the matter, which increases the likelihood of future financial statement restatements (Ma et al., 2023). A negative tone also signals credit risk. Negative descriptions related to working capital or cash flow may suggest potential issues with a company's solvency, indicating possible difficulties in meeting debt obligations and signaling credit risk (Camacho-Miñano et al., 2024; Kousenidis et al., 2019).

Creditors are likely to focus on the negative tone of KAMs. They often face an asymmetry between risk and return, with banks particularly concerned about downside risks when evaluating credit (Hasan et al., 2014). Additionally, KAMs typically involve complex, subjective, and challenging judgments during auditing (Lennox et al., 2023; Reid et al., 2019). Consequently, positive KAMs may be perceived as a form of "appeasement," potentially diminishing creditors' trust in audit information. In contrast, negative KAMs convey risk signals and enhance the trust of external users (Köhler et al., 2020). Therefore, creditors are more likely to consider the negative tone in KAMs.

A negative tone in KAMs can prompt banks to tighten their credit terms. Banks rely on various sources of information to assess a client's credit and information risk, which helps to determine differentiated credit terms, including pricing (e.g., loan interest rate) and non-pricing terms (e.g., loan amount, maturity, and collateral) (Bharath et al., 2008; Chen et al., 2016; Graham et al., 2008). When KAMs' tone becomes more negative, implying higher information and credit risk for a company, rational banks may respond by tightening loan contract terms, such as increasing interest rates, reducing loan amounts and durations, and increasing collateral requirements.

Accordingly, I propose the following hypothesis:

Hypothesis 1. *The more negative the tone of KAMs, the stricter the bank lending terms.*

3. Research design

3.1. Sample selection and data sources

Auditing Standard No. 1504 was implemented for A + H share-listed companies in 2017 and extended to other listed companies in 2018. I select A-share listed companies on the Shanghai and Shenzhen stock exchanges from 2016 to 2022 as the initial sample. After excluding financial and insurance companies, companies listed for less than a year, and those with missing data, I obtain the final sample of 23,631 loan-company-year observations. Owing to data availability, I identify 1025 observations of loan interest rates. The tone of KAMs is then calculated using the BERT model. Data on new loans, company characteristics, and financial indicators are obtained from the CSMAR database. To mitigate the impact of outliers, all continuous variables are winsorized at the top and bottom 1 %.

3.2. Definitions of key variables

3.2.1. Bank credit decisions (Bank loan)

Following the literature (Chen et al., 2016; Porumb et al., 2021), I measure bank credit decisions using both pricing and non-pricing terms. Pricing terms include the loan interest rate (*Interest Rate*), which represents the actual interest rate of each new loan. Non-pricing terms include the loan amount (*LnLoan*), which is the natural logarithm of the amount of each new loan; loan maturity (*LnTerm*), the natural logarithm of the term of each new loan; and loan collateral (*Collateral*), which equals 1 if the collateral is required, and 0 otherwise.

3.2.2. Negativity of the tone of KAMs (Tone)

Traditional dictionary methods, such as Loughran and McDonald's (2011) dictionary based on company 10-K reports, analyze texts by counting word frequencies and treating words as independent entities. This approach can misclassify terms such as "risk" and "misstatement" as purely negative. In auditing, these terms reflect auditors' risk assessments rather than emotional expressions, limiting the effectiveness of dictionary methods in analyzing the tone of KAMs (Seebeck and Kaya, 2023). In contrast, BERT, a transformer-based model with billions of parameters pre-trained on large corpora, effectively captures context and sentiment. Fine-tuned BERT models outperform traditional dictionaries and other machine-learning models in sentiment classification[1] (Huang et al., 2023; Kirtac and Germano, 2024). Therefore, I employ BERT to accurately classify the tones of KAMs.

Under the modern risk-oriented auditing model, KAMs represent some of the most complex, subjective, and challenging judgments made during the auditing process (Lennox et al., 2023; Reid et al., 2019). Unlike management's positive bias, auditors' risk perceptions are often reflected in KAMs' negative tone (Seebeck and Kaya, 2023; Smith, 2023). Following Huang et al. (2023), I construct a net negative-tone indicator using the BERT model. Table 1 provides the details, with higher tone values indicating stronger negativity.[2]

3.3. Model specification

Following existing research (Ertugrul et al., 2017; Porumb et al., 2021), I construct the following regression model:

$$Bank\ loan_{i,t+1} = \gamma_0 + \gamma_1 Tone_{i,t} + \sum Controls_{i,t} + \sum Direction + \sum Ind + \sum Year + \varepsilon_{i,t} \quad (1)$$

where the dependent variable $Bankloan_{i,t+1}$ represents bank loan decisions, measured by the loan interest rate (*Interest Rate*), loan amount (*LnLoan*), loan maturity (*LnTerm*), and loan collateral (*Collateral*). The independent variable $Tone_{i,t}$ represents the net negative

[1] Huang et al. (2023) demonstrate the considerable advantages of the fine-tuned BERT model in sentiment detection, achieving an accuracy of 85.0%. In comparison, the accuracy of other models is as follows: Naive Bayes, 73.6%; Support Vector Machine, 72.6%; Random Forest, 71.9%; Convolutional Neural Network, 75.1%; and Long Short-Term Memory network, 76.3%. All these values are lower than BERT's. Kirtac and Germano (2024) find that a BERT model fine-tuned with specific corpora achieves a 72.5% accuracy rate in sentiment detection, significantly outperforming the 50.11% accuracy of Loughran and McDonald's (2011) dictionary method. The BERT model also demonstrates advantages in precision and recall metrics.

[2] Huang et al. (2023) find that the BERT model has a considerable advantage in detecting negative sentiment, which has been validated in the sentiment classification of analyst reports and earnings conference calls. In my study, primarily owing to the standardized wording and structure of KAM texts, high-quality data labeling, and BERT's formidable contextual understanding, the fine-tuned BERT model achieves an overall accuracy of 98.8% in classifying negative, neutral, and positive sentiments.

Table 1
Constructing the net negative tone indicators for KAMs using the BERT model.

Step	Name	Description
Step 1	Pre-training	The pre-trained model used here is the Chinese version of BERT-base-Chinese released by Google, with 110 million parameters. The data used during the pre-training phase include the entire content of Chinese Wikipedia, totaling over 1.2 million entries and 550 million characters.
Step 2	Fine-tuning	I randomly extract 1000 texts of KAMs from 2016 to 2022. Subsequently, I split these texts into individual sentences and manually annotate each sentence for sentiment.[a] Finally, I randomly select 80 % and 20 % of these as the training and test sets, respectively.[b]
Step 3	Sentiment classification	I use the trained BERT model to classify the sentiment of KAMs for listed companies, counting the number of positive, neutral, and negative sentences in each KAM text.
Step 4	Indicator construction	Tone = (Number of negative sentences – Number of positive sentences) / Total number of sentences.

Note: This table details the steps involved in constructing the net negative tone index for KAMs using the BERT language model. This analysis focuses solely on the audit reports of A-share companies listed on the Shanghai and Shenzhen stock exchanges.

[a] A sentence is a natural unit for expressing ideas in language. The BERT algorithm accepts only 512 tokens as the input, which limits the size of the input text. Therefore, drawing on Huang et al. (2023), I categorize the sentiment of KAMs at the single-sentence level. Specifically, I divide the 1000 KAMs into individual sentences using periods and semicolons in the “matter description paragraph” and “audit response paragraph” of each KAM. Then, I label them with positive, neutral, or negative sentiments, categorizing and labeling each accordingly.

[b] I repeat the above training and testing steps 10 times. In the test set, the prediction accuracy of the fine-tuned BERT model exceeds 98 %, demonstrating its robustness and strong generalization capabilities.

Table 2
Definitions and descriptions of key variables.

Variable type	Variable name	Variable code	Variable definition
Dependent variables	Loan interest rate	*Interest Rate*	Actual interest rate (%) of each new loan
	Loan amount	*LnLoan*	Natural logarithm of the amount (in millions of CNY) of each new loan
	Loan maturity	*LnTerm*	Natural logarithm of the maturity (in years) of each new loan
	Loan collateral	*Collateral*	Equals 1 if collateral is required for a new loan, and 0 otherwise
Independent variable	Net negative tone	*Tone*	(Number of negative sentences – Number of positive sentences) / Total number of sentences in the text
Control variables	Firm size	*Size*	Natural logarithm of total assets at the end of the period
	Leverage ratio	*Lev*	Total liabilities / Total assets
	Return on assets	*Roa*	Net profit after tax / Total assets
	Accounts receivable ratio	*Rec*	Net accounts receivable / Total assets
	Other receivables ratio	*Other*	Net other receivables /Total assets
	Loss indicator	*Loss*	Equals 1 if net profit is less than 0 for the year, otherwise 0
	Listing age	*Age*	ln (current year – listing year + 1)
	Board size	*Board*	Natural logarithm of the number of board members
	Proportion of independent directors	*Indep*	Number of independent directors / Total number of directors
	Dual role	*Dual*	Equals 1 if the chairman and CEO are the same person, and 0 otherwise
	Ownership concentration	*Top1*	Shares held by the largest shareholder / Total shares
	Big Four auditor	*Big4*	Equals 1 if audited by a Big Four firm, and 0 otherwise
	Type of audit opinion	*Mao*	Equals 1 if the audit opinion is modified, and 0 otherwise
	Tone of management	*Mtone*	(Number of negative words – Number of positive words) / Total number of words in management text[a]
	Number of KAMs	*Num*	Total number of KAMs
	Length of KAMs	*Len*	Natural logarithm of the total number of words in KAMs
	Number of lenders	*Nlenders*	Number of banks participating in the loan
	Top five state-owned banks	*Bank5*	Equals 1 if the lending bank is ICBC, ABC, BOC, CCB, or BoCom, and 0 otherwise
	Loan purpose	*Direction*	Dummy variables based on the purpose of the loan
	Industry-fixed effects	*Ind*	Industry classification according to CSRC’s “Guidelines for the Industry Classification of Listed Companies”
	Year-fixed effects	*Year*	Year dummy variables

Note: This table provides definitions and coding of the key variables used in the analysis, covering the dependent, independent, and control variables and the fixed effects for loan purposes, industry, and year. I examine the impact of the tone KAMs’ on corporate loan terms.

[a] Following Ertugrul et al. (2017), I use the financial sentiment dictionary from Loughran and McDonald (2011) to evaluate managerial tone. The indicator construction involves three steps: first, the dictionary is translated into Chinese to suit the context of Chinese annual reports; second, the Jieba tool is employed to tokenize MD&A texts, filter stop words, and count positive and negative words; finally, the managerial tone indicator (*Mtone*) is calculated as (negative words - positive words) / total words. Robustness tests using alternative indicators, such as (negative words - positive words) / (negative words + positive words), and the “Chinese Sentiment Polarity Dictionary” from National Taiwan University confirm the consistency of findings, verifying their robustness.

Table 3
Descriptive statistics.

Variable	Sample size	Mean	Standard deviation	Median	Minimum value	Maximum value
Interest Rate	1025	5.274	1.717	4.785	1.350	10.700
LnLoan	23,631	4.927	1.310	4.942	1.792	8.243
LnTerm	23,631	0.320	0.622	0	−0.693	2.708
Collateral	23,631	0.200	0.400	0	0	1
Tone	23,631	0.112	0.050	0.118	−0.033	0.222
Size	23,631	23.000	1.392	22.910	20.320	26.640
Lev	23,631	0.522	0.183	0.537	0.120	0.916
Roa	23,631	0.027	0.063	0.031	−0.258	0.177
Rec	23,631	0.147	0.124	0.118	0.001	0.559
Other	23,631	0.020	0.029	0.010	0	0.158
Loss	23,631	0.147	0.354	0	0	1
Age	23,631	2.411	0.747	2.485	0	3.367
Board	23,631	2.128	0.211	2.197	1.609	2.708
Indep	23,631	0.378	0.053	0.364	0.333	0.571
Dual	23,631	0.264	0.441	0	0	1
Top1	23,631	0.318	0.147	0.293	0.079	0.687
Big4	23,631	0.075	0.264	0	0	1
Mao	23,631	0.025	0.155	0	0	1
Mtone	23,631	0.003	0.010	0.003	−0.019	0.027
Num	23,631	2.152	0.595	2	1	6
Len	23,631	7.118	0.372	7.147	5.919	7.950
Nlenders	23,631	1.025	0.396	1	1	27
Bank5	23,631	0.289	0.453	0	0	1

Note: This table provides descriptive statistics for key variables, including dependent variables, such as loan interest rate, loan amount, and loan term, along with the statistical characteristics of the net negative tone and control variables. Descriptive statistics include sample size, mean, standard deviation, median, and minimum and maximum values. For clarity, certain variables (e.g., loan amount and loan term) are presented in their natural logarithmic form. Actual values, such as loan amounts and terms, were computed as exp(x) from their logarithmic forms. For instance, the minimum values are approximately CNY 6.003 million and 0.5 years (or 180 days, assuming 360 days per year). Regression analysis was conducted using logarithmic transformations. All continuous variables were winsorized at the 1 % and 99 % levels to mitigate the influence of extreme values on the results.

tone of the KAMs for the current period. $Controls_{i,t}$ denotes the control variables, while *Direction, Ind*, and *Year* represent the fixed effects for loan purpose, industry, and year, respectively. $\varepsilon_{i,t}$ is the error term. To enhance the robustness of the regression results, standard errors are clustered at the firm level. I focus on γ_1, which indicates the relationship between KAMs' net negative tone and bank loan decisions. Table 2 provides detailed definitions of the relevant variables.

4. Empirical results

4.1. Descriptive statistical analysis

I use each new loan issued to a listed company as an observational unit. The results in Table 3 show that the average interest rate for new loans (*Interest Rate*) is 5.274 %, the mean logarithm of the loan amount (*LnLoan*) is 4.927 (approximately CNY 137.965 million), and the mean logarithm of the loan term (*LnTerm*) is 0.320 (approximately 1.377 years). Furthermore, the average collateral rate (*Collateral*) is 0.200, meaning that approximately 20.0 % of new loans require collateral. The mean net negative tone of KAMs (*Tone*) is 0.112, with a median of 0.118, ranging from -0.033 to 0.222, indicating significant variation in tone. The data for the control variables align with existing research, suggesting a reasonable data distribution.

4.2. Baseline regression

Table 4 presents the baseline regression results. In Column (1), the coefficient for loan interest rate (*Interest Rate*) is 3.166, which is significantly positive at the 1 % level.[3] This indicates that a 1-standard deviation increase in *Tone* corresponds to a 3.002 % increase in the average interest rate.[4] Column (2) shows a negative coefficient of −1.445 for loan amount (*LnLoan*), which implies a 7.225 %

[3] In China's emerging market, characterized by the coexistence of policy-driven and market-oriented mechanisms, zero-interest loans often originate from policy support (e.g., government-led industrial promotion or regional development initiatives) or intra-group support (e.g., interest-free loans from parent companies to subsidiaries). Loan spread, defined as the difference between the nominal interest rate and market benchmark rate, eliminates the influence of policy interventions and internal support on interest rates, providing a more accurate reflection of the risk-pricing logic in the credit market. I employ loan spread as an alternative measure to verify the robustness of the baseline regression conclusions. The relevant results are available upon request.

[4] Based on computation adopted from Ertugrul et al. (2017), the economic significance for the loan interest rate is 3.002%, which matches the result of 3.166 × 0.050/5.274 = 3.002%.

Table 4
Tone of KAMs and bank loan decisions.

	(1)	(2)	(3)	(4)
	Interest Rate	*LnLoan*	*LnTerm*	*Collateral*
Tone	3.166***	−1.445***	−1.697***	9.429***
	(2.915)	(−2.963)	(−6.439)	(5.261)
Size	−0.164*	0.289***	0.025*	−0.165**
	(−1.859)	(7.651)	(1.906)	(−2.027)
Lev	1.843***	0.291	0.210***	0.990*
	(2.981)	(1.561)	(2.934)	(1.957)
Roa	−4.366**	1.336***	−0.084	−1.289
	(−2.038)	(2.940)	(−0.504)	(−1.088)
Rec	−0.086	−0.998***	−0.113	1.134*
	(−0.091)	(−3.651)	(−0.740)	(1.691)
Other	7.193***	−2.781**	−1.041**	−0.770
	(2.653)	(−2.272)	(−2.235)	(−0.237)
Loss	−0.551**	0.056	−0.065**	−0.230
	(−2.071)	(0.731)	(−2.223)	(−1.214)
Age	−0.189	−0.038	−0.035**	−0.302**
	(−1.550)	(−0.756)	(−2.090)	(−2.480)
Board	−0.709	0.076	−0.012	−0.830*
	(−1.514)	(0.322)	(−0.140)	(−1.885)
Indep	−2.448*	−0.314	−0.252	−1.805
	(−1.680)	(−0.465)	(−1.034)	(−1.231)
Dual	0.227	0.103*	0.017	0.275*
	(1.363)	(1.732)	(0.812)	(1.960)
Top1	−2.220***	0.255	−0.163	−1.806**
	(−3.699)	(1.115)	(−1.479)	(−2.472)
Big4	−0.259	0.116	0.138**	0.468
	(−0.824)	(0.852)	(1.970)	(0.995)
Mao	0.621*	0.153	0.055	0.513**
	(1.738)	(1.545)	(1.351)	(2.117)
Mtone	10.589	−4.398	−2.093	12.733
	(1.197)	(−1.259)	(−1.526)	(1.603)
Num	0.078	−0.039	0.051*	0.274*
	(0.413)	(−0.636)	(1.846)	(1.832)
Len	0.168	−0.010	−0.078*	−0.384
	(0.625)	(−0.095)	(−1.923)	(−1.465)
Nlenders	−0.602***	0.223***	0.042***	−0.019
	(−3.802)	(5.597)	(2.588)	(−0.322)
Bank5	−0.410***	0.099**	0.061***	−0.134
	(−2.699)	(2.423)	(3.781)	(−1.512)
Direction / Ind / Year	Yes	Yes	Yes	Yes
Constant	11.395***	−1.961	0.517	4.944**
	(5.027)	(−1.549)	(1.344)	(2.008)
N	1025	23,631	23,631	23,631
R^2 / *Pseudo* R^2	0.450	0.181	0.321	0.142

Note: This table presents the baseline regression results (Hypothesis 1). When the dependent variable is *Collateral*, a logit model is used, with *z*-values in parentheses, and a *Pseudo* R^2 provided. For the other dependent variables, ordinary least squares (OLS) regression is applied, with *t*-values in parentheses and company-level clustered standard errors. R^2 values are reported for these regressions. Table 2 provides detailed descriptions of all variables. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

decrease (approximately CNY 9.970 million) for each standard deviation increase in *Tone.*[5] In Column (3), the loan term (*LnTerm*) coefficient is −1.697, reflecting an 8.485 % reduction in the loan term.[6] Given an average loan term of 1.377 years, this corresponds to a decrease of approximately 0.117 years (or roughly 42 days, based on a 360-day year) for a standard deviation increase of 1 in *Tone.*[7]

[5] The economic significance for the loan amount is $-1.445 \times 0.050 = -7.225\%$. As the mean Ln*Loan* value is 4.927, corresponding to an average loan amount of approximately CNY 137.965 million, the estimated impact is a reduction of 7.225% or approximately CNY 9.970 million.

[6] Based on the reviewer's suggestion, I conduct supplementary analyses using a count model. Specifically, the loan maturity data, originally measured in years, are converted into integer values in days (i.e., multiplying by 360 and rounding to the nearest whole number) to satisfy the count model's assumptions. Given the over-dispersion observed in the loan maturity data (variance > mean), I employ a negative binomial regression model for the analysis. The results demonstrate that the direction and significance of the regression coefficients for the core variables are consistent with those of the baseline regression, further confirming the robustness of the conclusions. Owing to space limitations, the detailed results are not included in the main text but can be provided upon request.

[7] The economic significance for the loan term is approximately $-1.697 \times 0.050 = -8.485\%$. With an average loan term (*LnTerm*) of 1.377 years, this implies a reduction of about 0.117 years. Assuming a year is defined as 360 days under the international financial convention, this corresponds to a decrease of approximately 42 days ($0.117 \times 360 = 42.120$, rounded to the nearest whole number).

Table 5
Instrumental variable approach.

	(1)	(2)	(3)	(4)	(5)
	Tone	*Interest Rate*	*LnLoan*	*LnTerm*	*Collateral*
Aqi	0.009***				
	(6.580)				
Tone		56.505*	–14.563***	–4.069**	9.937**
		(1.720)	(–3.410)	(–2.200)	(2.160)
Control	Yes	Yes	Yes	Yes	Yes
N	21,766	906	21,766	21,766	21,766
First-stage *F*-value		77.350			
(*p*-value)		0.000			

Note: This table presents the regression results obtained using the instrumental variable (IV) approach IV. For the dependent variable *Collateral*, a logit model is applied, with *z*-values reported in parentheses, and a *Pseudo* R^2 is provided. For all other dependent variables, OLS regression is applied, with t-values in parentheses and company-level clustered standard errors reported alongside the R^2 values. The regression includes all control variables from the baseline regression as well as fixed effects for loan purpose, industry, and year. Table 2 provides detailed descriptions of all variables. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Column (4) shows a positive coefficient of 9.429 for collateral (*Collateral*), indicating a 71.600 % higher probability of collateral under a strong negative tone.[8] These results support Hypothesis 1 as banks raise interest rates, reduce loan amounts, shorten terms, and increase collateral requirements in response to risks.

The results show that KAMs' tone has a more substantial signaling effect in China's credit market. Audit opinion (*Mao*) and the number of KAMs (*Num*) impact credit decisions, which is consistent with the existing literature (Chen et al., 2016; Porumb et al., 2021); however, their influence is weaker than that of the tone conveyed in the KAMs. This may be because audit opinions and the number of KAMs are more homogeneous, while auditors' discretion in wording allows KAMs' tone to better reflect risk (Seebeck and Kaya, 2023; Smith, 2023). Building on Ertugrul et al.'s (2017) finding that the tone of annual reports is related to stricter credit terms, my study shows that management's tone (*Mtone*) does not significantly affect when both audit and management tones coexist. Therefore, KAMs are considered more credible than management, leading lenders to trust auditors' professional judgments. This preference reinforces the signaling effect of KAMs' tone, aligning with experimental findings (Christensen et al., 2014; Sirois et al., 2018).

4.3. Robustness checks

4.3.1. Instrumental variable approach

I employ an instrumental variable approach to address endogeneity concerns, using air pollution (*AQI*) in the location of the listed company during the audit period as an instrument for *Tone*.[9] Table 5 reports a first-stage *F*-value of 77.350, which is statistically significant at the 1 % level and indicates that the instrument is relevant. Additionally, *Tone* shows a significant association with bank loan decisions at the 10 % level, at minimum, thereby confirming the robustness of the baseline regression results.

4.3.2. Replacing the measurement of the independent variable

In the baseline regression, I construct a net negative-tone indicator for KAMs using the BERT model. Additionally, I reconstruct another indicator (*Tone1*) employing the dictionary method.[10] The results presented in Columns (1)–(4) of Table 6 are significant at the 10 % level, thereby confirming the robustness of the baseline regression results.

[8] To provide a more intuitive explanation of economic significance and reduce the complexity of interpreting nonlinear relationships, I divide the tone variable into "strongly negative" and "weakly negative" groups based on whether it exceeds the sample median and use these as dummy variables in the regression analysis. The logit regression results show a coefficient of 0.540, which, when converted into an odds ratio, indicates that the probability of requiring collateral is 71.600% higher under the strongly negative scenario than under the weakly negative scenario. This approach enhances the clarity and interpretability of economic significance.

[9] Following Chen et al. (2022), I calculate the distance between the company's location and the nearest air pollution monitoring station and use the Air Quality Index from that station, scaled by dividing by 100, as a measure of air pollution. The air pollution data are averaged over a 28-day window prior to the auditor's signing date, with higher values indicating more severe pollution. This instrument satisfies both the relevance and exogeneity requirements. Finance research shows that air pollution significantly amplifies pessimism among market participants, such as investors and analysts (Lepori, 2016; Dong et al., 2021). More direct evidence comes from Chen et al. (2021), who find that air pollution during the audit period directly affects auditors' mood, making them more cautious in their risk assessment. Therefore, I expect air pollution during the audit period to lead to a more negative tone in KAMs, which supports the instrument's relevance. Simultaneously, air pollution does not directly influence bank loan decisions, thereby meeting the exogeneity condition.

[10] First, the financial sentiment dictionary of Loughran and McDonald (2011) is used as a base, with manual screening and expansion to account for differences between Chinese and English. Second, the Jieba segmentation tool is used to tokenize the text and remove stop words, and positive and negative words are counted. Finally, I define *Tone1* as the ratio of the difference between the number of negative and positive words to their sum and use it as an alternative measure of *Tone* in the regression model.

Table 6
Reconstructed indicators based on the dictionary method.

	(1)	(2)	(3)	(4)
	Interest Rate	*LnLoan*	*LnTerm*	*Collateral*
Tone1	0.547*	−0.354***	−0.120**	0.820*
	(1.769)	(−2.638)	(−2.404)	(1.682)
Control	Yes	Yes	Yes	Yes
Direction / Ind / Year	Yes	Yes	Yes	Yes
Constant	13.830***	−2.753**	−0.202	10.461***
	(5.543)	(−2.341)	(−0.539)	(4.178)
N	961	21,282	21,282	21,282
R^2 / *Pseudo* R^2	0.449	0.188	0.310	0.124

Note: This table presents the regression results obtained using the dictionary method to re-measure the explanatory variables. For the dependent variable *Collateral*, a logit model is applied with *z*-values reported in parentheses, and the *Pseudo* R^2 provided. For the other dependent variables, ordinary least squares regression is used, with *t*-values in parentheses, company-level clustered standard errors, and R^2 values reported. The regression includes all control variables from the baseline regression and fixed effects for loan purposes, industry, and year. Table 2 provides detailed descriptions of all variables. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 7
Mechanism analysis results.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Interest Rate	*LnLoan*	*LnTerm*	*Collateral*	*Interest Rate*	*LnLoan*	*LnTerm*	*Collateral*
Tone	7.364***	−2.926***	−3.118***	13.904***	8.918***	−3.960***	−3.252***	15.836***
	(3.742)	(−4.195)	(−6.951)	(6.078)	(2.844)	(−3.182)	(−4.180)	(4.467)
Inform	−0.226	−0.131	−0.222***	0.235				
	(−0.927)	(−1.348)	(−3.699)	(0.657)				
Tone × *Inform*	−4.446**	1.770**	1.750***	−5.521*				
	(−2.114)	(2.152)	(3.867)	(−1.909)				
Credit					−0.313	−0.309***	−0.220**	0.971**
					(−0.799)	(−2.786)	(−2.375)	(2.116)
Tone × *Credit*					−5.876*	2.781**	1.719**	−7.246*
					(−1.897)	(2.254)	(2.249)	(−1.926)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dire / Ind / Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	10.871***	−1.840	0.699*	4.781*	11.386***	−1.620	0.756*	3.951
	(4.877)	(−1.454)	(1.846)	(1.938)	(5.112)	(−1.252)	(1.951)	(1.565)
N	1025	23,631	23,631	23,631	1025	23,631	23,631	23,631
R^2 / *Pseudo* R^2	0.467	0.181	0.324	0.146	0.471	0.182	0.323	0.143

Note: This table presents the results of mechanistic analyzes. When the dependent variable is *Collateral*, a logit model is applied with *z*-values reported in parentheses, and a *Pseudo* R^2 provided. For other dependent variables, ordinary least squares regression is used, with *t*-values in parentheses and company-level clustered standard errors reported along with R^2 values. The regression includes all control variables from the baseline regression and fixed effects for loan purposes, industry, and year. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

5. Mechanism analysis

As information recipients, banks adjust their lending decisions by interpreting signals. The stronger the negative tone, the greater the bank's perceived information and credit risk, leading to tighter credit terms. I empirically examine whether the tone of KAMs helps reveal information risk and assess credit risk. First, I evaluate the quality of corporate information disclosure to measure information risk. Specifically, I define a dummy variable (*Inform*) that equals 1 if the information disclosure rating is "excellent" or "good," and 0 otherwise.[11] Next, I introduce the interaction term (*Tone* × *Inform*) into the baseline regression model. The results, presented in Columns (1)–(4) of Table 7, show that the interaction term is significant at least at the 10 % level and has a coefficient opposite in sign to that of *Tone* in the baseline regression. This confirms the presence of the information risk revelation effect. Second, I use the average ratings from third-party credit rating agencies to measure corporate credit risk. Specifically, I set a dummy variable (*Credit*) to 1 if the average credit rating is above the industry median, and 0 otherwise.[12] I introduce the interaction term (*Tone* × *Credit*) into the baseline regression model. The results, shown in Columns (5)–(8) of Table 7, indicate that the interaction term is significant at least at the 10 %

[11] Each year, the exchange evaluates listed companies on six indicators of the quality of accounting information disclosure: truthfulness, accuracy, completeness, timeliness, fairness, and compliance. The assessments are categorized into four levels: excellent, good, qualified, and unqualified, with each successive level indicating a decline in disclosure quality and increase in information risk.

[12] Credit rating agencies assign companies ratings across nine levels: C, CC, CCC, B, BB, BBB, A, AA, and AAA. I sequentially assign values from 1 to 9 to these ratings and calculate the average credit rating issued by the credit rating agencies for each company annually. A higher average value indicates lower credit risk.

Table 8
Tone-type analysis results.

	(1) *Interest Rate*	(2) *LnLoan*	(3) *LnTerm*	(4) *Collateral*	(5) *Interest Rate*	(6) *LnLoan*	(7) *LnTerm*	(8) *Collateral*
Postone	−5.413**	2.275***	4.573***	−3.822**				
	(−2.560)	(3.155)	(10.050)	(−2.048)				
Negtone	2.490**	−1.112*	−0.747***	11.130***				
	(2.111)	(−1.909)	(−2.734)	(4.977)				
Abtone					2.732**	−0.800	−0.753***	7.289***
					(2.258)	(−1.302)	(−2.769)	(3.107)
Ntone					5.282**	−2.465***	−3.498***	7.675***
					(2.491)	(−3.151)	(−7.319)	(2.678)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dire / Ind / Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	11.579***	−2.047	0.282	4.750*	10.403***	−2.188*	0.865**	5.456**
	(5.160)	(−1.624)	(0.744)	(1.924)	(4.259)	(−1.709)	(2.077)	(2.172)
N	1025	23,631	23,631	23,631	1015	22,540	22,540	22,540
R^2 / *Pseudo* R^2	0.451	0.181	0.336	0.144	0.448	0.174	0.330	0.133

Note: This table presents the results of the tone-type heterogeneity analysis. When the dependent variable is *Collateral*, a logit model is used with *z*-values in parentheses and the *Pseudo* R^2 provided. For other dependent variables, OLS regression is applied, with *t*-values in parentheses and company-level clustered standard errors alongside the R^2 values. The regression includes all control variables from the baseline regression and fixed effects for loan purposes, industry, and year. The construction of the *Postone* and *Negtone* indicators is detailed in footnote 13, while the *Abtone* and *Ntone* indicators are described in Appendix A. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

level and has a coefficient opposite in sign to that of *Tone* in the baseline regression. This confirms the existence of the credit risk assessment effect.

6. Heterogeneity analysis

To investigate the heterogeneous effects of signal types, following Lennox et al. (2023), I categorize the tones of KAMs into positive (*Postone*), negative (*Negtone*),[13] expected net negative (*Ntone*), and unexpected net negative (*Abtone*) tones.[14] These tones are also included separately in the regression equation. The results in Table 8 indicate that a positive tone (*Postone*) has a loosening effect on bank loan decisions, while a negative tone (*Negtone*) has a tightening effect, which demonstrates asymmetry in their impacts. Additionally, the expected net negative tone (*Ntone*) exerts a more significant influence on loan decisions, exhibiting a stronger signaling effect than the unexpected net negative tone (*Abtone*).

7. Conclusion

KAMs' negative tone reflects auditors' strong perceptions of clients' financial risks and is considered more credible than management's tone, thereby serving as a substantial risk signal for bank credit decisions. This influence arises from the ability of KAMs' tone to help banks reveal information risk and effectively evaluate credit risk. Furthermore, negative and positive signals affect credit decisions asymmetrically, with expected net negative tones exerting a stronger signaling effect than unexpected ones. These findings align with signaling theory, which posits that in an information-asymmetric environment, banks rely more on credible audit reports to make credit decisions. Overall, my study provides theoretical and empirical insights into the role of audit information in the debt market.

Research involving human participants and/or animals statement

Not applicable.

Informed consent statement

Not applicable.

CRediT authorship contribution statement

Mingli Duan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

[13] I define positive tone (*Postone*) as the ratio of the number of positive sentences to the total number of sentences in the KAMs' text and negative tone (*Negtone*) as the ratio of the number of negative sentences to the total number of sentences in the KAMs' text.
[14] The specific calculation method can be found in Appendix A.

Conflict of interest statement

None declared.

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Appendix A. Methodology for breaking down the tone of KAMs

Following prior studies (Huang et al., 2014; Lennox et al., 2023), I construct the following model:

$$\begin{aligned} Tone_{i,t} = {} & \rho_0 + \rho_1 Size_{i,t} + \rho_2 Lev_{i,t} + \rho_3 Inv_{i,t} + \rho_4 Fixed_{i,t} + \rho_5 Groth_{i,t} + \rho_6 Problem_{i,t} + \rho_7 Loss_{i,t} + \rho_8 Bm_{i,t} + \rho_9 Gc_{i,t} + \rho_{10} Def_{i,t} \\ & + \rho_{11} Goodwill_{i,t} + \rho_{12} Acq_{i,t} + \rho_{13} Subnum_{i,t} + \rho_{14} Analysts_{i,t} + \rho_{15} Kamnum_{i,t} + \rho_{16} Kamlen_{i,t} + \sum Accfirm + \sum Ind \\ & + \sum Year + \varepsilon_{i,t} \end{aligned}$$

where $Size_{i,t}$ represents the size of the company; $Lev_{i,t}$ indicates financial leverage; $Inv_{i,t}$ measures the ratio of inventory to total assets; $Fixed_{i,t}$ denotes the ratio of fixed assets to total assets; and $Groth_{i,t}$ is the revenue growth rate. $Problem_{i,t}$ identifies whether financial restatements or accounting irregularities occurred in the previous two years. $Loss_{i,t}$ indicates whether the company recorded a financial loss in year t. $Bm_{i,t}$ measures the book-to-market ratio. $Gc_{i,t}$ specifies whether a going concern is mentioned in KAMs or audit opinions. $Def_{i,t}$ represents the ratio of income tax expenses to total assets. $Goodwill_{i,t}$ represents the goodwill-to-total assets ratio. $Acq_{i,t}$ is the ratio of mergers and acquisitions (M&A) to total assets. $Subnum_{i,t}$ is the natural logarithm of the number of subsidiaries. $Analysts_{i,t}$ measures the level of analyst attention. $Kamnum_{i,t}$ indicates the number of key audit issues and $Kamlen_{i,t}$ denotes the natural logarithm of the number of words in KAMs. The accounting firm, industry, and year fixed effects are *Accfirm, Ind*, and *Year*, respectively. The model predicts the normal (*Ntone*) and abnormal negative tones (*Abtone*) of the KAMs.

Data availability

Data will be made available on request.

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